

Cloud Computing

Presented by: Dr M Chere



CLOUD COMPUTING:
Master the Concepts, Architecture, and Applications with Real-world examples and Case studies

Kamal Kant Hiran Ruchi Doshi Dr. Temitayo Fagbola Mehul Mahrish

Semester Test 1

- **Date** : Monday 13th April 2026
- **Venue**: Lecture Hall C
- **Time**: 11:00 12:30

Lesson 4: Cloud Computing Virtualization Technology

Presented by: Dr M Chere



CLOUD COMPUTING:

Master the Concepts, Architecture, and Applications with
Real-world examples and Case studies

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"Virtual Reality is better than the 'limited' real world."

Mark Zuckerberg, Co-founder, Facebook

Objectives



To learn about the **basic concepts of virtualization**



To learn about the **classification of virtualization** based on **techniques** and **usage**



To know the **role of virtualization** in **building an efficient cloud architecture**

Content



Understanding virtualization



Adopting virtualization



Techniques of virtualization



How virtualization works



Types of virtualization



Virtualization in the cloud

Understanding Virtualisation

Introduction

- Virtualization technologies
 - Are the **backbone** to the **successful adoption** of the **cloud computing paradigm**
- Virtualization not only **upgrades the relationship** between **hardware** and **software** but also helps **utilize their full capabilities**

Understanding Virtualization

Virtualization

- Technology that enables us to **create** a logical/ virtual object of an actual physical object
- It **involves** the **distribution of capabilities** of a **physical machine** among many users or environments, **imitated** using software

Significance

- **Enhanced resource sharing** by many users and **improved computer performance** in terms of resource utilization and application flexibility:
 - Reduced costs
 - Scalability
 - Disaster recovery

Resources to Virtualise

Can virtualize resources such as:

- Operating system
- Storage
- Computer network

Understanding Virtualization

Virtual Machine

Is the application software that is responsible for creating these virtual instances

The end users have the same look and feel on a VM as they would have on the actual physical hardware

These VMs are:

- Portable
- Platform-independent
- Sandboxed from the host system

Hypervisor

A Virtual Machine Monitor (VMM) used to properly manage the working of VMs and maintain the integrity of virtual environments

It can be a software agent, a firmware, or a hardware device

The physical machine on which the VMM runs is called the host machine, and each VM is termed as a guest Machine

Understanding Virtualization

Hypervisor

- Software component that **manages** multiple virtual machines in a computer
- It **ensures** that each **virtual machine** gets the **allocated resources** and does not **interfere with the operation** of other virtual machines
- There are two types of **hypervisors**

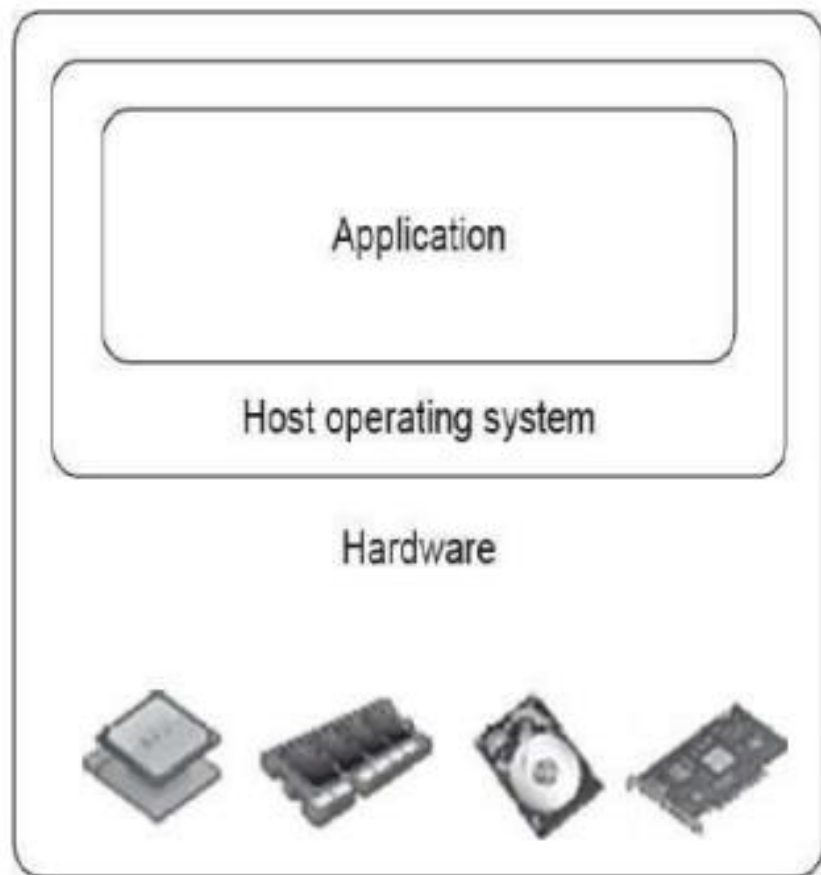
Types of Hypervisor

Type 1 hypervisor (**bare-metal hypervisor**)

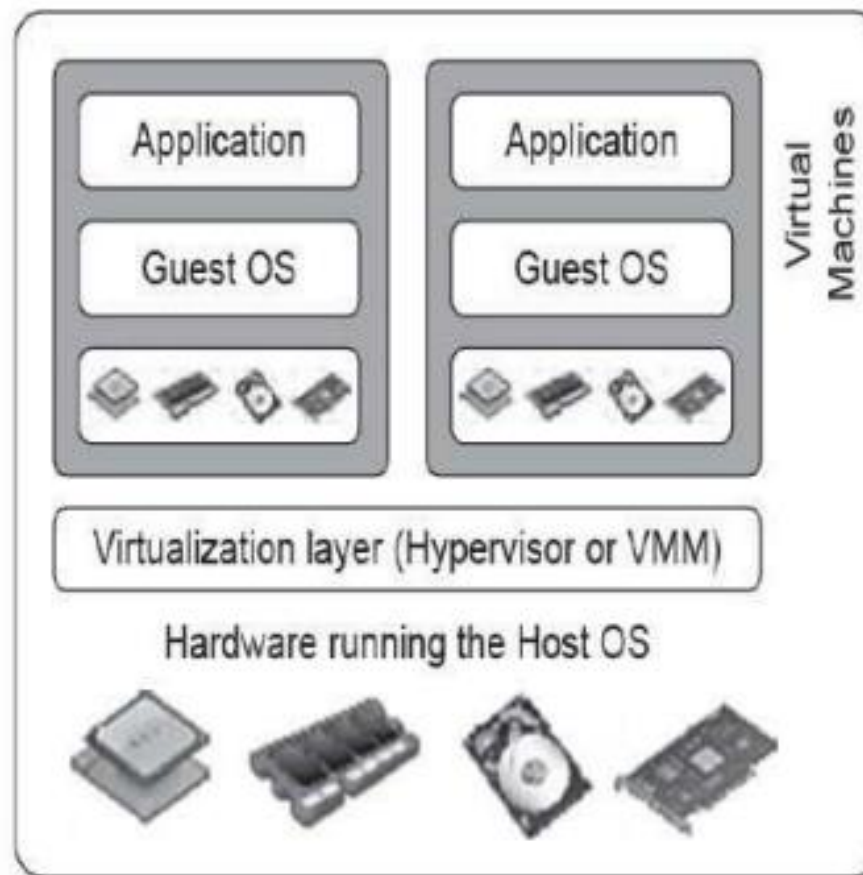
- It is a **hypervisor program installed directly** on the computer's hardware **instead** of the **operating system**.
- Have **better performance** and are **commonly** used by **enterprise applications**
- **Kernel-based Virtual Machine (KVM)** uses the **type 1 hypervisor** to host multiple virtual machines on the Linux operating system

Type 2 hypervisor (**hosted**)

- Known as a **hosted hypervisor**
- Is installed on an **operating system**
- Suitable for end-user computing



(a) Traditional computer



(b) After virtualization

A traditional computer runs with a host operating system specially tailored for its hardware architecture. After virtualization, different user applications managed by their operating systems (guest OS) can run on the same hardware, independent of the host OS. This is often done by adding additional software, called a virtualization layer.

Adopting Virtualization

Adopting Virtualization

- Various organizations are adopting virtualization due to various factors:

Cost Reduction	Increased Ubiquity	Easier backup and Redeployment	Environment Protection
Limiting hardware resources, thereby reducing significant operational and maintenance costs , which ultimately reduces the electricity, air conditioning, and cooling costs	Decreased downtime - by increasing isolated instances of resources, and thereby increasing efficiency, uptime, and business continuity	Through simplified backup procedures not only on virtualized servers but also on virtual machines which can move from one server to another quickly	By safeguarding planet Earth , saving energy, and reducing e-waste generation

Challenges of Virtualization Adoption

- Virtual Machines (VMs) can become **uncontrollable** and create a **virtual bottleneck** that can cause **disasters** in the performance of a virtualized system – require **proper management**
- **Virtual backups** are the most **prominent IT challenges** for the business community and can be very **tricky at times**, which **threaten** to **compromise the performance** of the whole system

Challenges of Virtualization Adoption

- **Additional** and **unanticipated costs** often result when **specialized hardware** and **licenses** are required to keep **high availability** and **performance**, e.g.
 - **Burden of VM management**
 - **Storage area network requirements**
 - **Increased headcount**
 - **Security vulnerabilities and compliance risks**

Challenges of Virtualization Adoption

Over-worked resources occur due to VM and network card saturation, and can result in performance issues like reduced bandwidth, bogging of I/O intensive operations

Techniques of Virtualization

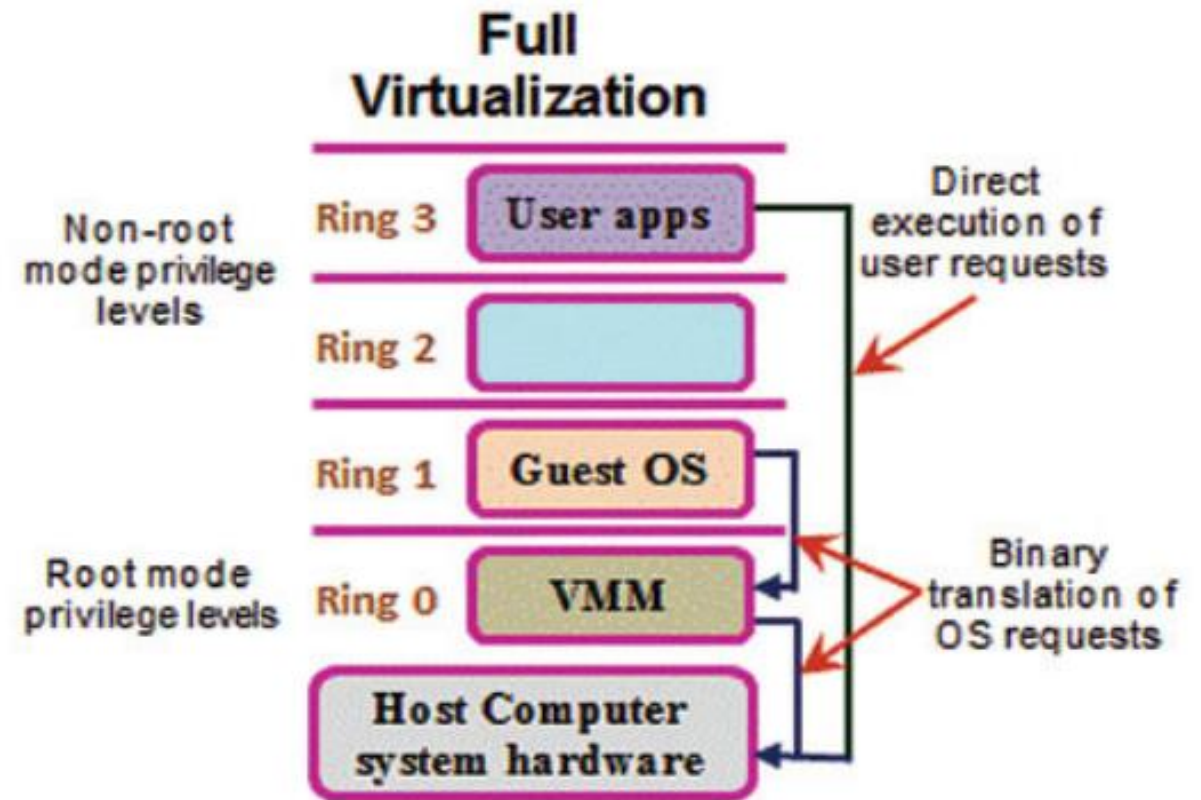
- Virtualization techniques can be **categorized** into three categories:
 - Full Virtualization
 - Para Virtualization
 - Hardware-assisted Virtualization

Full Virtualization

- Refers to a virtualization technique that allows for **complete simulation** of the underlying hardware in a virtual machine
- Allows multiple operating systems (OS) to **run concurrently** on a single physical machine, each operating as if it has its dedicated hardware
 - In **full virtualization**, primary hardware is replicated and made available to the guest operating system
 - The **guest OS** is not aware that it is being virtualized and requires no further modification

Full virtualization

In **full virtualization**, primary hardware is replicated and made available to the guest operating system. The **guest OS** is not aware that it is being virtualized and requires no **further modification**, but the **user-level** code (executed in Ring 3) is **directly executed** on the **processor** for **high-performance virtualization**. Since this technique translates kernel code to replace nonvirtualizable instructions with new sequences of instructions that have the intended effect on the virtual hardware, the VMM must be executed in Ring 0



Full virtualization

- **Full Virtualized architecture** provides **virtualized memory, virtual devices, and virtual BIOS**.
- To run the **guest operating system** without any modification, a **particular technique** is used, which is known as **Binary Translation**
- **Examples of full virtualization:**
 - Microsoft Hyper-V – Integrated with Windows Server for hybrid cloud deployments,
 - VMWare vSphere/ESXi
 - KVM (Kernel-based Virtual Machine) – Open-source virtualization for Linux

Para Virtualization

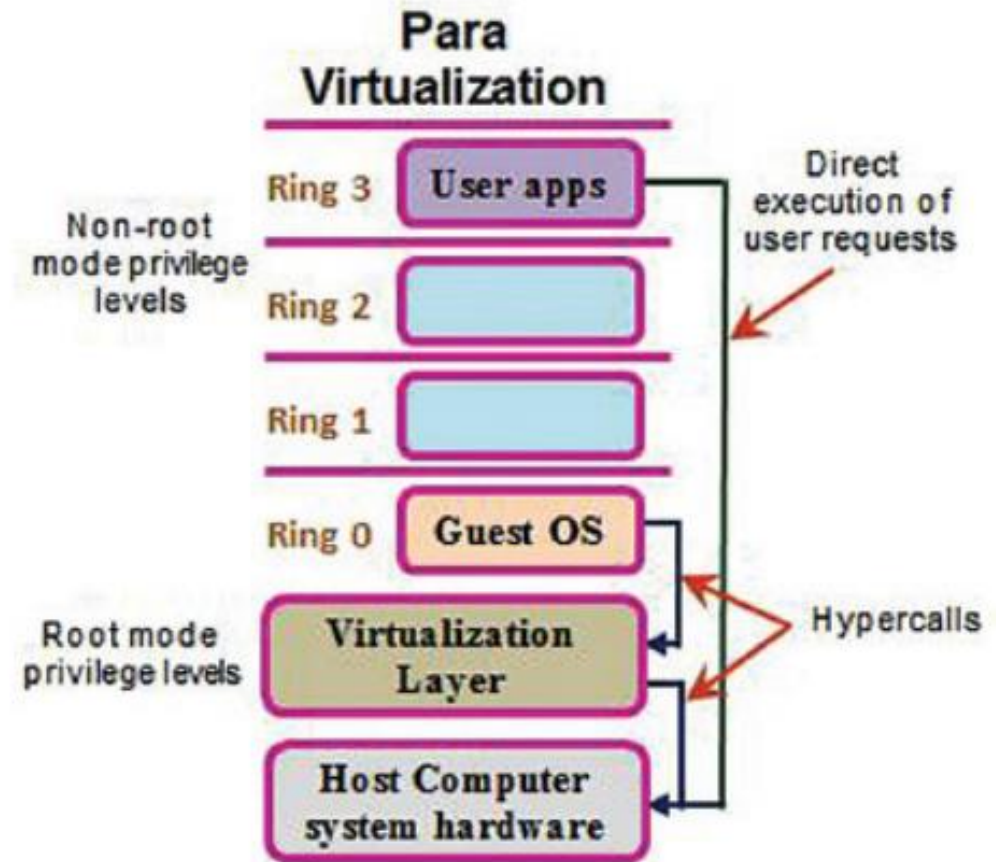
- Refers to the Partial hardware simulation requiring OS modification:
 - Also known as OS-assisted Virtualization
- Improves the communication between the guest operating system and the hypervisor by replacing the non-virtualizable instructions with hypercalls that communicate directly with the virtualization layer hypervisor

Para Virtualization

- **Examples include:**
 - **Xen Hypervisor:** An open-source Type 1 (bare-metal) hypervisor that supports paravirtualization (e.g. Amazon Web Services)
 - **VMware ESXi:** A leading enterprise-level Type 1 hypervisor supporting both full and paravirtualization (e.g., Data centers, enterprise environments for better performance and scalability)
 - **Oracle VM Server:** Oracle's Type 1 hypervisor, based on the Xen hypervisor, supports paravirtualization (e.g. runs Oracle databases and applications in optimized virtual environments)

Para Virtualization

Para Virtualization is an extension of virtualization that recompiles the guest operating system installing it inside a virtual machine. It is sometimes also known as OS-assisted Virtualization. Paravirtualization improves the communication between the guest operating system and the hypervisor by replacing the non-virtualizable instructions with hypercalls that communicate directly with the virtualization layer hypervisor. Due to this reason, the OS must run its privileged instructions in **Ring 0**. **Ring 3** is for less privileged user applications. As paravirtualization modifies operating systems before executing, its compatibility and portability are poor. The open-source Xen project is an example of paravirtualization that virtualizes the processor and memory using a modified Linux kernel and virtualizes the I/O using custom guest OS device drivers.

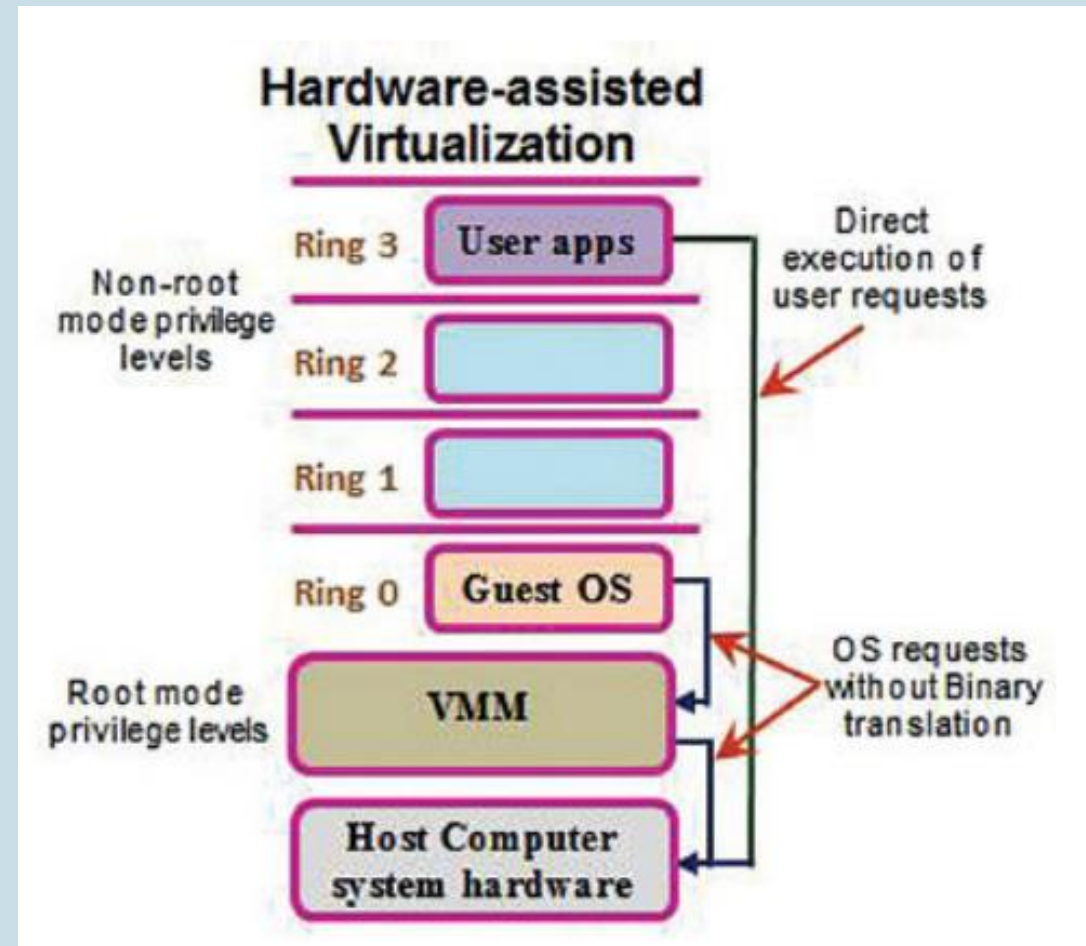


Hardware-assisted Virtualization

- As the name suggests, it is a **virtualization approach** that **provides full virtualization using hardware capabilities**
- The **technology is invented by Intel and AMD** to improve the **performance of processor utilization** and to **overcome other challenges like memory address resolution and instruction translation**

Hardware-assisted virtualization

- Hardware virtualization is the embedding of a VM into the hardware component of a server. The idea is to aggregate small physical servers into one large physical server and use the processor effectively. Since the OS needs direct access to hardware and memory modules in this approach, it must execute its privileged instructions in Ring 0. While user-level applications typically run in Ring 3. Privileged instructions are executed into a CPU execution mode that allows the VMM to run in a new root mode below Ring 0



How Virtualization Works?

How Virtualization Works

- Goal of virtualization technology:
 - To **create** an independent environment for different applications on a single hardware machine
- This is done by creating virtual instances of **operating systems, applications**, etc., designed to run directly on hardware.
- The technique **extends the capabilities** of your machine and **allows** you to **run multiple applications** (especially operating systems) at the same time over a single hardware configuration

How Virtualization Works

- Virtual Machine (VM)
 - Is the **application software** that is **responsible** for creating these virtual instances
 - The **end users** have the **same look and feel** on a **VM** as they would have on the **actual physical** hardware
 - These **VMs** are:
 - Portable
 - Platform-independent
 - Sandboxed from the host system

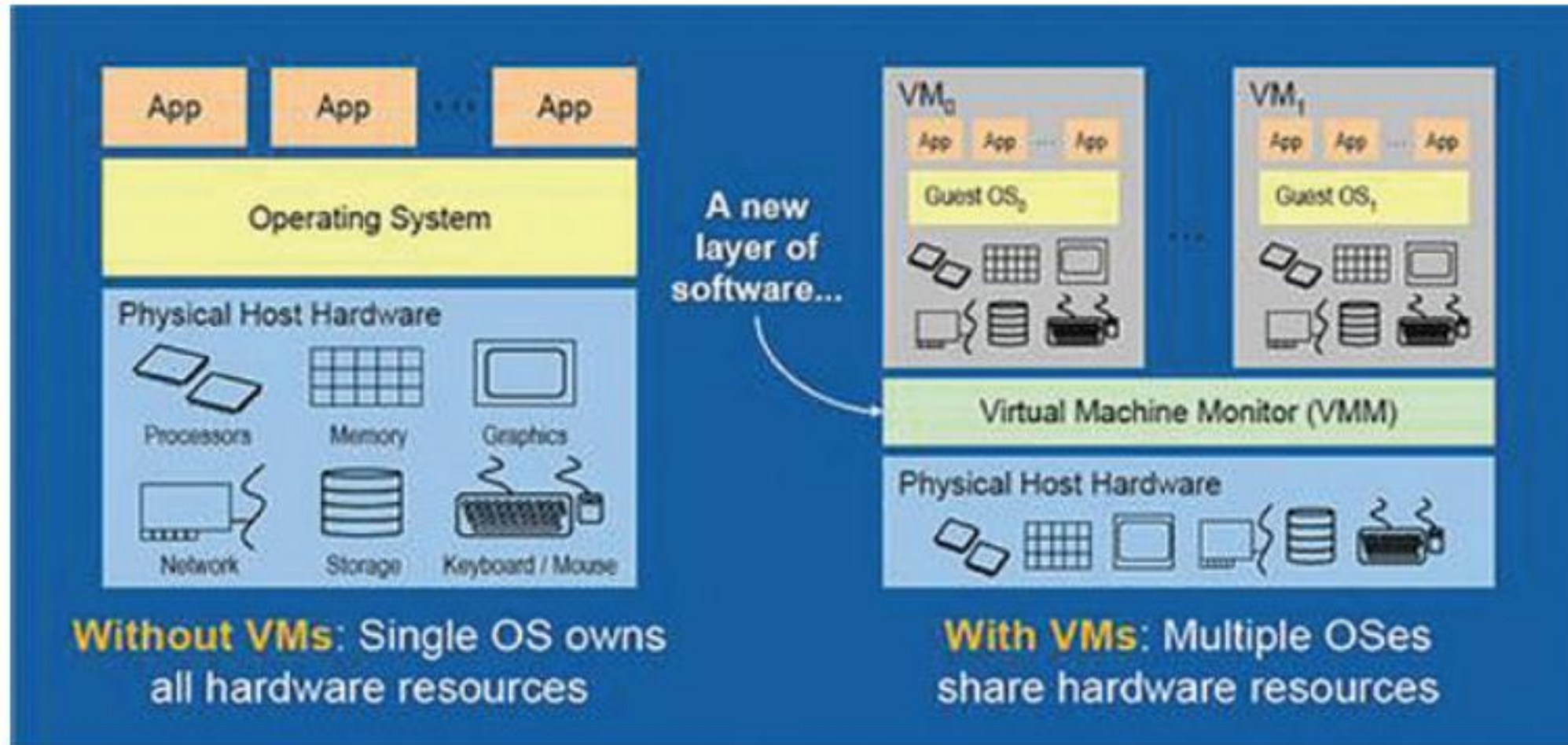
How Virtualization Works

- A Virtual Machine (VM)
 - A host can even run multiple VMs simultaneously on a single hardware configuration
 - But to create a VM we need substantial processing power, physical memory, and network bandwidth

How Virtualization Works

- To properly manage the working of VMs and maintain the integrity of virtual environments, a **Virtual Machine Monitor** (VMM) or **Hypervisor** is used
- It can be a **software agent**, a **firmware**, or a **hardware device**.
- The **physical machine** on which the **VMM** runs is called the **host machine** and each **VM** is termed a **guest machine**

Virtual architecture

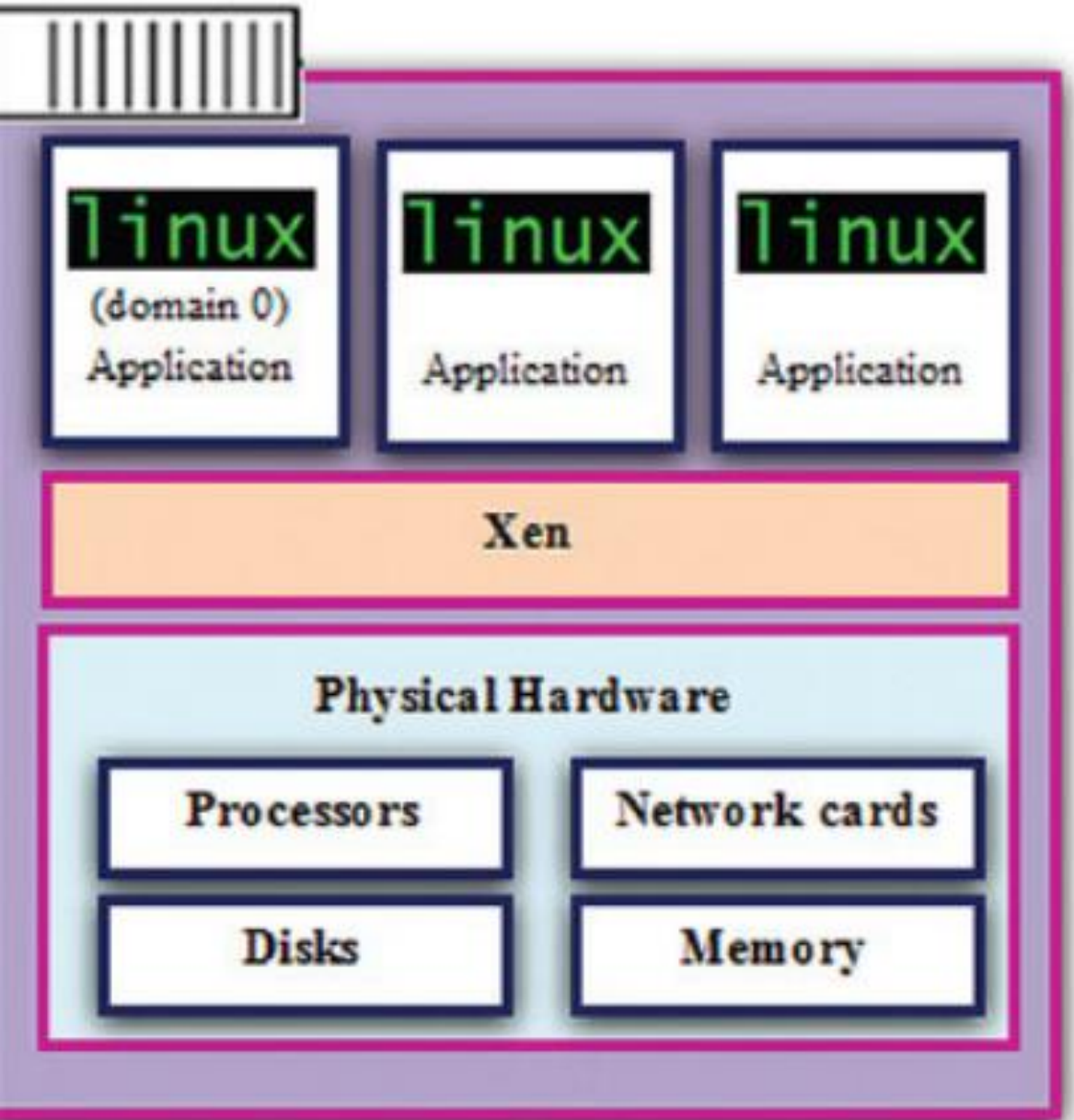


Popular Virtual Machine Monitor VMM/Hypervisors

ZEN- Executive Engineer

- Referred to as a **free and open-source type-1 hypervisor**, developed by the Xen Project, used in the AWS Cloud, server virtualization, Infrastructure-as-a-Service (IaaS), desktop virtualization, security virtualization
- **Allows multiple operating systems** to run concurrently on the same hardware
- **Known for its flexibility** in virtualization applications, powering clouds and other environment

Xen hypervisor architecture



- Shows the **Xen hypervisor architecture**
- The **hypervisor layer** is directly installed on Physical Hardware
- There is no need to pre-install any host operating system to run virtual machines

XEN Features

Memory ballooning

CPU pools

Remus fault tolerance

Virtual machine introspection

XEN Features

Memory ballooning

- An **advanced memory management technique** in which the hypervisor can claim the unused memory from one guest machine and share it with the other guest machines within a host
- Allows the **amount of RAM required by guest VMs** to exceed the actual amount of physical RAM available on the host

CPU pools

- Is a **unique resource management** feature of XEN version 4.2.
- This technique **divides the physical cores on the machine** into **different pools**, each with its customizable CPU scheduler
- At runtime, a **VM is assigned to one of the pools**, but it can further migrate between any other pools through the course of its execution
- Since the **scheduler is customizable**, requests can be made for different scheduling parameters for different VMs

XEN Features

Remus Fault Tolerance

- It is **responsible** for the **high availability** of **Xen**
- This is done by **recurrently creating live backups** of running **VMs** to the **backup server** which **automatically activates** in case of **failure**

Virtual Machine Introspection

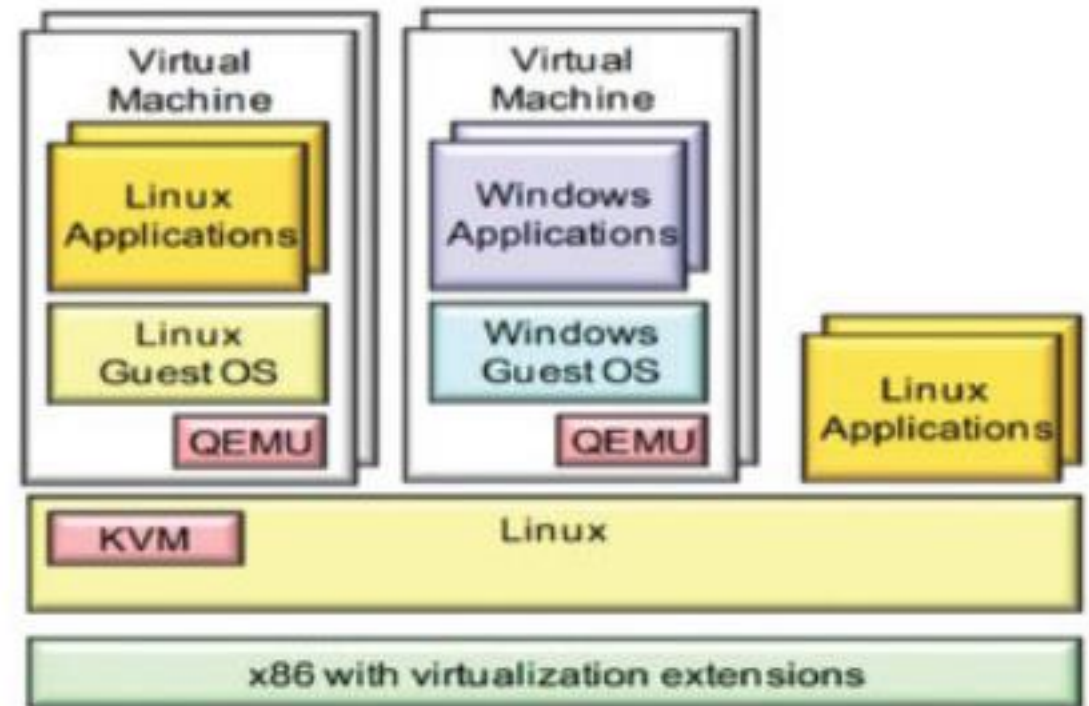
- It is a **security technique** in **Xen** that **audits** the **sensitive memory areas** of guest machines using specialized hardware support with minimal overhead

Kernel-based Virtual Machine (KVM)

- KVM is an open-source virtualization layer fused into the mainline Linux kernel
- It converts a Linux-operated machine into a Type 1, bare-metal hypervisor that can run multiple but isolated virtual environments
- Since KVM is integrated into Linux Kernel, modules like memory management, CPU scheduling, input/output (I/O), and device management, are already built-in.
- Each VM is treated as a standard Linux process which is scheduled by a typical Linux Scheduler with virtualized hardware

KVM Architecture

- Shows a typical KVM architecture
- The KVM module is embedded with the Linux operating system, over which virtual machines run simultaneously with standalone Linux applications



Kernel-based Virtual Machine Features

- **Mandatory Access Control (MAC) security:**
 - Mechanisms are implemented for the security of guest machines
 - VM Security and VM Isolation are provided using enhanced Linux features, namely, **security-enhanced Linux (SELinux)** and **secure virtualization (sVirt)**

Kernel-based Virtual Machine Features

- Hardware and storage support:
 - KVM is compatible with a wide variety of Linux-certified hardware platforms, local disks, and network-attached storage (NAS)

Kernel-based Virtual Machine Features

- Live VM migration:
 - It is a feature of KVM through which running VMs can be migrated between hosts without service interruption

QEMU

Is an **open-source emulator** for hardware virtualization

It stands for **Quick Emulator** and acts as a **virtual machine monitor** when executed using the KVM kernel module in Linux

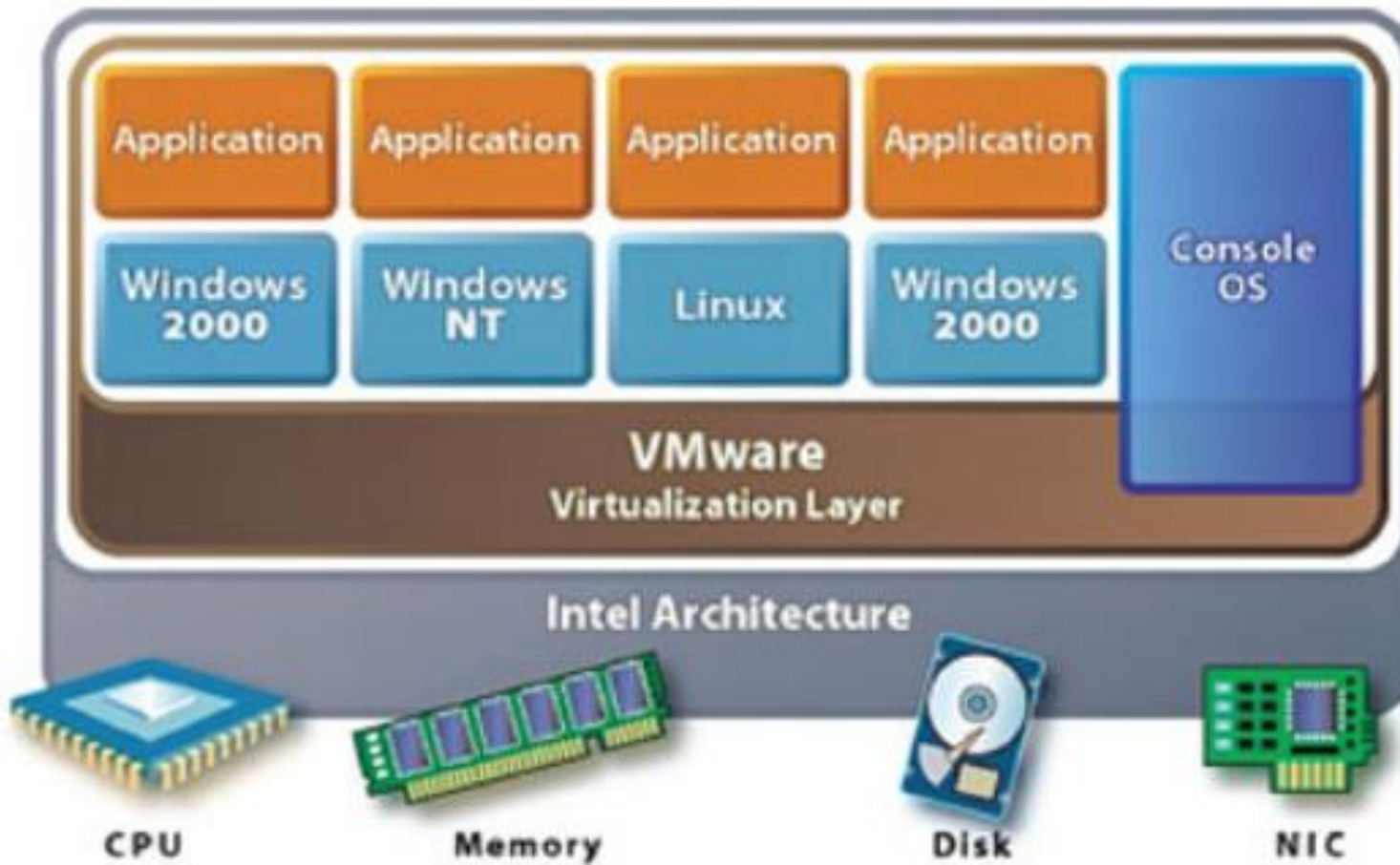
VMware

- **VMware** (now a subsidiary of Dell Technologies) is a **virtualization software provider** based in Palo Alto, California
- The company has gained its position among the key virtualization providers in the industry

VMware

- VMware classified its products into two categories:
 - **Desktop applications:** Desktop applications are compatible with almost all operating systems and provide three major applications which are as follows:
 - **VMware workstation:** It is a virtualized software package in which multiple instances of operating systems (either similar or different) are installed on a single hardware machine.

VMware Architecture



Shows the architecture of VMware

It requires a **console OS** to be installed over the **hardware**. It creates a **software-based virtualization layer** over which multiple instances of operating systems can be hosted simultaneously

All the instances of the operating systems can be similar or different, but they share a single hardware configuration

Popular Virtual Machine Monitor VMM/Hypervisors

- For more please refer to the textbook, or other materials on the following hypervisors and many more:
 - VirtualBox
 - Citrix

Types of Virtualization

Types of Virtualization



In cloud computing, there are various types including:

- Server
- Storage
- Network
- Application
- Desktop

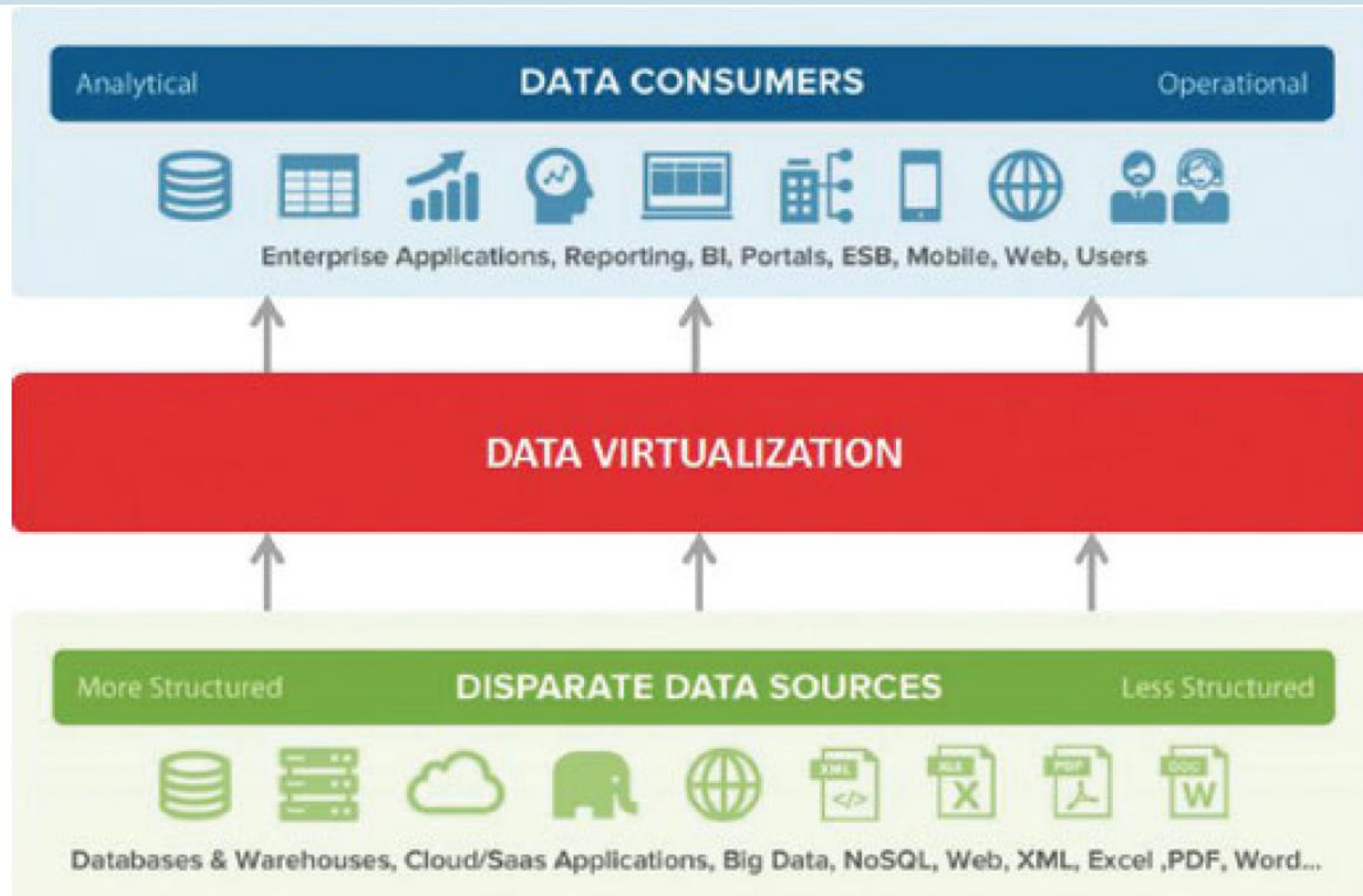


To enable **efficient resource utilization** and **flexible infrastructure management**

Types of Virtualization

- Data virtualization
 - **Availability** of the **right data** at the **right time** - main objectives of virtualization
 - Data virtualization is analogous to data agility in which an application is allowed to access data **irrespective** of its **technical details, formatting style,** and **physical location**

Types of Virtualization



- Data virtualization aligns different sources into a **single virtual data layer** that provides **unified access and integration data service**.
- Tools like Red Hat's JBoss, Denodo, etc. fetch data from **multiple heterogeneous sources**, integrate it, and transform it as per the user's need.
- This ultimately results in **faster access to data, less replication, and high data agility**

Features of Data Virtualization

- Modern data integration:
 - The **integration** and **transformation** of data is similar to the traditional **Extract-Transform-Load Model (ETL Model)** but leverages modern features like delivery of **real-time data**, **data federation**, and **data agility**

Desktop Virtualization

- Desktop virtualization is a technique in which the **virtualization layer** runs on **top of a hypervisor** and **provides desktop provisioning**
- It is also alternatively called a **virtual desktop infrastructure (VDI)**
- We can relate this **type of virtualization** with the **traditional client-server model** where a **client requests for service** from a **centralized and remotely located server**
- In desktop virtualization, **VDI** is **responsible** for **hosting the desktop environment** in a **VM** that runs on a **centralized or remote server**
- This is the reason why we also sometimes refer to **desktop virtualization** as **client virtualization**

Desktop Virtualization

Shows a **Virtual Desktop**, which is a **two-facet centralized server**

User Experience (Thick Client, Desktop, Laptop) on one side and applications (OS, Provisioning & Update, User Data & Personalization, Application Virtualization) on the other side

The **remote server** is responsible for **disaster recovery, security, and availability** and **backup of data**



Features of Desktop Virtualization

The user's computer dies; the user's desktop does not die

- If hardware damage occurs, the failed hardware can be quickly replaced and simply reconnected to the virtual desktop

No installations or updates

- Since the operating system and other application software are centrally managed, there is no overhead of installation and regular version updates

Inter device compatibility

- The virtual desktop can be accessed by either a desktop machine or by a tablet or mobile device
- Multiple users can share a common virtual desktop environment, that is, the desktop or application needs to be installed only once, and it will be available to multiple users

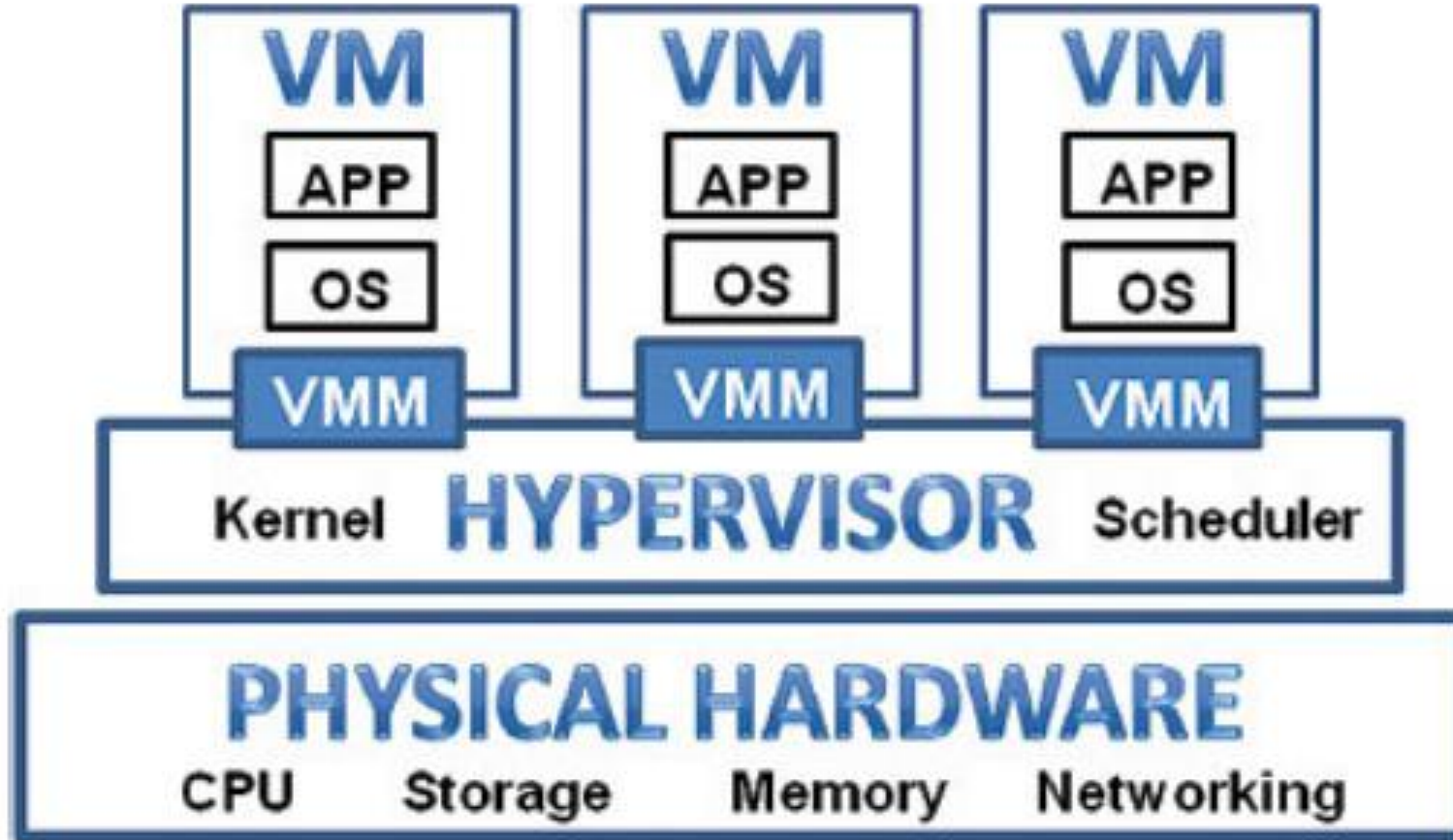
CPU Virtualization

- CPU virtualization allows a single processor to act as multiple individual CPUs, that is, two separate systems running into a single machine
- The objective of CPU virtualization is to enable the users to run different operating systems simultaneously
- Multiple virtual machine monitors are installed over the hypervisor to allow the execution of multiple instances of operating systems

CPU Virtualization

- Prominent CPUs, including **Intel** (Intel Virtualization Technology or Intel VT) and **AMD** (AMD-V) support **CPU virtualization**
- Since **virtualization** sometimes requires **kernel level** and **control sensitive instructions** which tend to change **memory mapping** and **resource configuration**, CPU virtualization is generally **disabled by default** in BIOS and needs to be **enabled manually** once using the CPU configuration settings

CPU Virtualization



Features of CPU Virtualization

Virtual processor IDs (VPID)	Descriptor table exiting	Pause-loop exiting	Extended page table (EPT)
- These are the unique IDs given to each VM currently running - This prevents the CPU from flushing out the data structures from the transition look-aside buffers (TLBs) at the time of context-switching VMs - VPIDs therefore help achieve flexibility and quality of service in terms of live VM migration	This feature prevents the relocation of key system data structures and thereby protects the guest OS from internal vulnerabilities	This feature enables to detect spin locks in the guest software and avoids lock-holder preemption which reduces overhead and improves performance	This technique allows the capability of the guest OS to handle page faults and modify its page tables

Network Virtualization

- In network virtualization, hardware and software network resources and their functionalities are compressed into a software-based administrative entity
- The ultimate result is a virtual network that is highly efficient in terms of utilization with less time overhead

Network Virtualization



Network virtualization **creates** the virtualized combination of available resources by **splitting up** the **bandwidth** into **channels** so that each device or user can have shared access to all the resources on the network

Classification of Network Virtualization

Two categories

Internal network virtualization

- This type of virtualization uses a **single system to act as a hypervisor** (Xen/KVM) to control **virtualized network interface cards** (VNCs)
- Each host can have one or more NICs and each NIC can be a base for multiple VNICs

External network virtualization

- In this type of virtualization, a **single virtual network** is **created** by either a **combination** or **division of multiple local area networks**, administered by the **software system**. **VLAN** and **network switch** are the components of this type of virtualization

Network Virtualization Features



Isolation:

This feature is kept among various logical **network partitions** to **avoid any kind of interference** and **reduce the performance**



Abstraction:

This feature **hides** the **underlying complexities** and **characteristics of network elements** from **applications** and **users**

Storage Virtualization

- Is the **assembling of physical storage** from **heterogeneous storage devices** to form a large **pool of memory** that is managed centrally
 - That is; a process of presenting a logical view of the physical storage resources to a host computer system, treating all storage media (hard disk, optical disk, tape, etc.) in the enterprise as a single pool of storage
- By allowing the **storage** to participate in **storage area networks (SANs)**, we are **increasing** the **efficiency, flexibility, and load balancing** of storage devices

Storage Virtualization



Network attached storage (NAS)

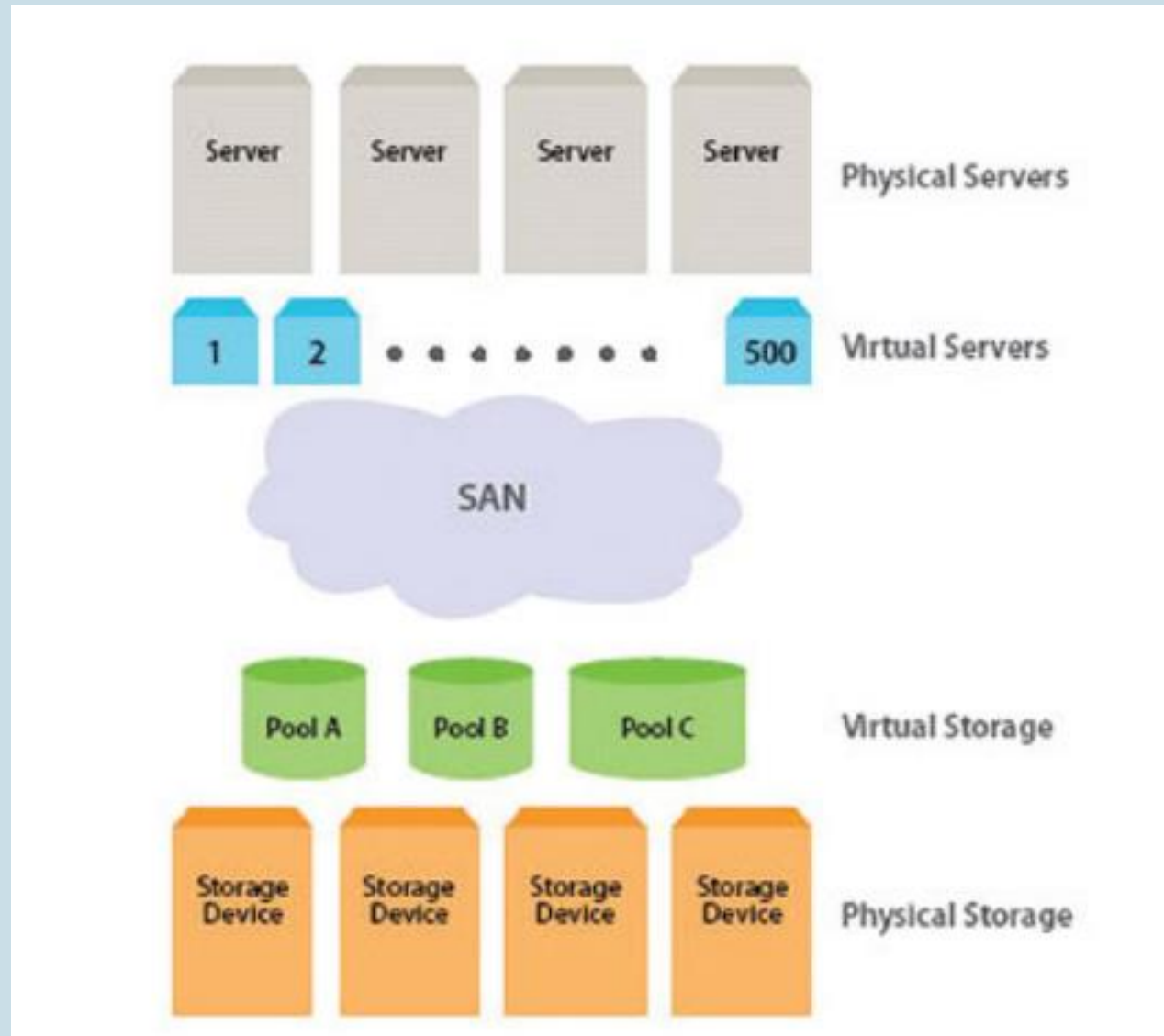
Either a **single device** or a **server**



Storage area networks (SAN)

A network of storage devices

Storage Virtualization



Features of Storage Virtualization



Non-disruptive data migration: This feature allows data to migrate without disturbing the concurrent I/O access



Improved utilization: Utilization is increased by pooling and migration



Thin provisioning: Technology **dynamically allocates** storage capacity to a **volume as per the usage requirement**, that is, it allows you to **tell the application that it has sufficient storage** without actually assigning storage to it

Server Virtualization

- In the traditional client-server architecture, the server machine runs only one instance of resources
- These resources can be processors, operating systems, application software, memory
- The idea of server virtualization is to divide a physical server machine into several logically isolated virtual machines and thereby create several instances of resources

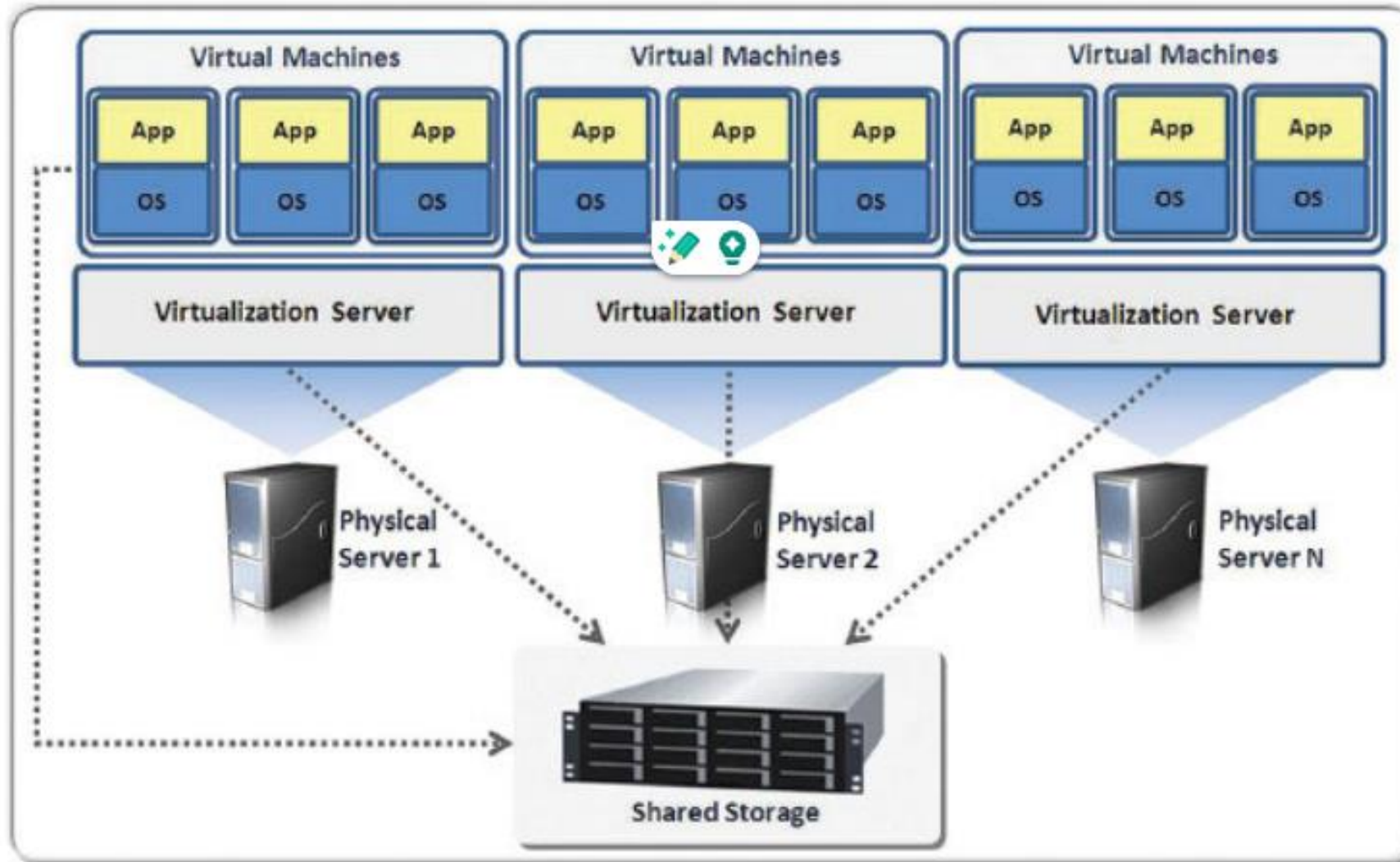
Server Virtualization

- **Adopting this approach** serves the **advantage** that rather than **deploying several servers** that **may not be fully utilized**, numerous **virtual machines** can run on the same **physical platform**
- For example, in a **company payroll system**, rather than using separate **servers for employee database**, email server, and document maintenance, all these applications can be virtualized onto a single server machine

Server Virtualization Techniques

Hypervisor or a VMM	Paravirtualization	Hardware virtualization	OS virtualization
A hypervisor or a VMM is a software layer that exists between the hardware machine and the operating system and is used to: • Handle kernel-level instructions • Queuing • Processing client requests	• Is a hypervisor-based virtualization in which the performance of a virtual machine is enhanced by pre-compilation of the guest OS before installing it on the virtual machine	• Is similar to paravirtualization except that some hardware assistance is provided • AMD—V Pacifica and Intel VT Vanderpool are examples of hardware-supported virtualization	▪ Provides multiple but logically isolated virtual machines that run on the operating system kernel ▪ The technique is also called the shared kernel approach because all the virtual machines share the same kernel of the host operating system. FreeVPS, and Linux Vserver are examples of OS level virtualization

Server Virtualization



Features of Server Virtualization

- **Partitioning:**
 - This feature allows multiple virtual servers to run on one physical server at the same time
- **Isolation:**
 - The virtual servers running on the physical server are completely isolated and don't affect the execution of each other
- **Encapsulation:**
 - All the information on virtual servers, including boot disks is saved in the file format

Features of Server Virtualization

- **Hardware independence:**
 - A virtual server runs as it is after migration to different hardware platforms.
- **Improved business continuity:**
 - Live migration of virtual servers to another physical server results in maintenance of servers without shutting down and hence improves availability and business continuity

Virtualization in Cloud

- Cloud technology aims to be :
 - ❑ Ubiquitous
 - ❑ Pervasive
 - ❑ Agile but with an unquenchable thirst for:
 - Collaboration
 - Sharing of resources
- Virtualization plays a very **vital role** in achieving these objectives and **empowering cloud**

Virtualization in Cloud

- To convert ideas into a streamlined business, enterprises need business applications that are very expensive and come with a complicated software stack
- Whenever a new version is released, the updating can cause incompatibility among the stack and break the whole system down
- To overcome this problem, virtualization is preferred

Virtualization in Cloud



Virtualization not only **provides** the **sharing of data** but also the **sharing of infrastructure**



Through **virtualization**, the **resources** become massively **scalable** and can be **integrated with IT-related capabilities**

Virtualization in Cloud

- The **capabilities of virtualization** and sharing of infrastructure in the Cloud computing paradigm are fulfilled by **three major distribution models**:
 - SAAS
 - PAAS
 - IAAS

Virtualization in Cloud

- Some **third-party providers** that host **applications, platforms, and infrastructure** over the **cloud** include:
 - Google Apps and Cisco WebEx for **SAAS**
 - Microsoft Azure, and Google App Engine for **PAAS**
 - Rackspace and Amazon AWS for **IAAS**

Summary

- The **cloud computing paradigm** is based on the concepts of virtualization, multi-tenancy, and shared infrastructure which is fulfilled by three major distribution models, namely, **SaaS**, **PaaS**, and **IaaS**

Next Lesson: Service Oriented Architecture